

Bilal Shihab

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PROFESSIONAL SUMMARY

Biomedical engineering student who builds on-device machine learning for healthcare, from iPhone apps to microcontroller firmware. Founder of a stealth-stage clinical AI startup, with experience training, quantizing and deploying models on EEG, EMG, ECG and clinical language data.

EDUCATION

UNIVERSITY OF TEXAS AT AUSTIN

Austin, TX

Bachelor of Science in Biomedical Engineering, Computational Track

May 2027

- **Relevant Coursework:** Embedded Systems, Real-Time Digital Signal Processing, Statistics, Differential Equations & Linear Algebra, Numerical Methods, Intro to Computational Engineering Design, Systems & Signals, Biomedical Instrumentation

SKILLS

- **Languages:** Python, MATLAB, C++, C, ARMv6-M Assembly, R
- **ML & Inference:** PyTorch, llama.cpp, quantization, ONNX, ExecuTorch, TensorFlow Lite, Metal GPU inference, Optuna, Captum
- **Infrastructure & Embedded:** Docker, Google Cloud Platform, Git, Raspberry Pi, TI MSPM0, UART

EXPERIENCE

WANG LABORATORY, UT AUSTIN

Austin, TX

Machine Learning Researcher

Feb 2026 - Present

- Built a Python signal-processing pipeline for a 12-subject gut-brain study: automated artifact removal, spectral and phase-amplitude coupling features from EEG/EGG recordings.
- Preparing the resulting feature set for ML models that predict optimal timing for targeted ultrasound vagus nerve stimulation.

LONGHORN NEUROTECH

Austin, TX

Edge AI Team Lead

Sep 2025 - Present

- Directed a team of student engineers to optimize neural networks for deployment on embedded systems (Raspberry Pi series).
- Designing a reusable quantization pipeline using ExecuTorch and an inference backend to support keyword spotting and neuroprosthetic control applications.
- Applied static [INT8 quantization and ONNX](#) export to an MLP classifying hand gestures from 8-channel EMG, reaching 0.43 ms/sample on a Raspberry Pi Zero 2 W against a 1.04 ms FP32 baseline, at a 0.13-point accuracy cost.

AI/ML Developer

Sep 2024 - Aug 2025

- Evaluated [CNN and Capsule Network](#) architectures for EEG motor-imagery classification, running Optuna searches over kernel sizes, layer depth, and dropout, and using Captum LayerGradCam for attribution analysis.
- Rearranged EEG channels to match scalp topography so the capsule network's spatial pooling grouped adjacent electrodes.

FUNCTIONAL OPTICAL IMAGING LAB, UT AUSTIN

Austin, TX

Researcher and Programmer

Sep 2023 - May 2025

- Developed a real-time [Laser Speckle UI](#) in Python, using Basler cameras and Arduinos for automated laser intensity control.
- Automated exposure and laser-current control against a center-weighted saturation limit, removing the need for a technician to adjust the system by hand.

TECHNICAL PROJECTS

STEALTH CLINICAL AI STARTUP (FOUNDER)

Jun 2026 - Present

- Founded a stealth-stage clinical AI startup and built its [iOS app](#), which records and analyzes clinical conversations entirely on the phone; no audio or transcript leaves the device.
- Benchmarked models on 240 labeled decisions and deployed a 4-bit Qwen 3.5-4B that beat a 7B model (85% vs 79%) with a 30% smaller download, running in under 500 MB of RAM.
- Developed a second-pass verifier that raised the model's agreement with hand-labeled grades from 65% to 84% on synthetic conversations, and cut per-evaluation latency ~60% (18.6 s → 7.3 s) by reusing the transcript's cached KV state.

LOCALABS - ON-DEVICE LAB REPORT TRANSLATOR

Apr 2026 - Present

- Built an [iOS app](#) that translates clinical lab reports into plain language using MedGemma 4B, Google's medical-domain model, running fully on-device via llama.cpp on Metal GPU; no cloud, no account, no telemetry.
- Designed an Apple Vision OCR → LLM pipeline producing five structured sections per report (summary, glossary, diet, questions, medications) in 12s on iPhone 17 from a 2.5 GB quantized model.

CLOUD TO EDGE HEART RATE MONITOR

Dec 2025 - Present

- Architected a [cloud-to-edge pipeline](#) that streams MIT-BIH ECG recordings from Google Cloud Storage to a Raspberry Pi 5, which relays them to a Raspberry Pi Pico in real time.
- Built a custom C inference engine on the Pico's dual-core Cortex-M0+ that classifies each heartbeat as normal, supraventricular ectopic or ventricular ectopic, with the second core running a fail-safe watchdog.

ECG PCB DESIGN

Mar 2025 - Apr 2025

- Built an [ECG board](#) with two teammates, designing the schematic and soldering: an AD620 instrumentation amplifier with high- and low-pass filtering that recorded a clean ECG in LabVIEW.